Film-Level and Musical Drivers of Soundtrack Streaming Performance

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I. Introduction and motivation

1.1 Background and motivation

Our team consists of music and film buffs, so focusing our project on movie soundtracks felt like a natural intersection of our interests. In the 90s and 2000s, we were avid collectors of soundtrack albums. Although film scoring has evolved, we still view scores as powerful elements to film's visceral appeal, and composers as central creative contributors. Our analysis focuses on 2015–2025, a period when film distribution became increasingly streaming-driven. Film discovery shifted to digital platforms, theatrical windows shortened, and long-tail consumption became standard – trends further accelerated by COVID-19. These changes raise questions like, "What drives sustained soundtrack listenership once films move into streaming ecosystems?"

Despite extensive research on music popularity, soundtracks remain understudied. Prior work has examined long-term listening patterns [2], the predictive value of audio features for mainstream music popularity [1,3,7], and listening behavior and recommendation systems [4,6]. This literature, however, largely focuses on individual tracks and digitally consumed mainstream music. We found only one related study on film soundtracks, which analyzes relationships between film scripts and musical attributes without addressing streaming engagement [5].

1.2 Questions

Soundtracks differ from the music most commonly studied. They are often instrumental-dominant, album-oriented, and revisited long after a film's release. Listening is frequently driven by kinship to the film itself: its popularity, awards, or independent appeal, rather than by playlist discovery. It is therefore unclear whether findings from all of music research apply to soundtracks. Motivated by this gap, we investigate how soundtrack listening relates to film- and album-level attributes during the modern streaming era. We asked whether listenership coincides with factors like film popularity, revenue, genre, and recognition, and whether film-related or musical attributes are more strongly associated with streaming outcomes. We hope this work provides insight for soundtrack producers, composers and fans, and shows the rewards of joining together cross-domain datasets for the study of media consumption behavior.

II. Data sources

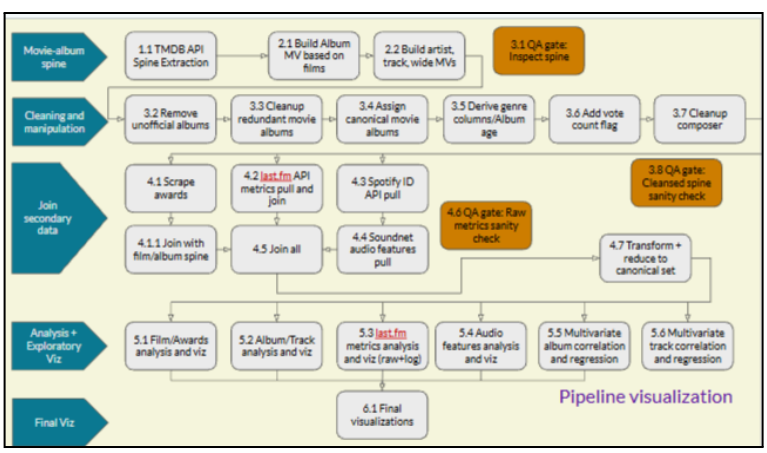

Figure 1. *Notebook-level (staging) pipeline with grouped stages and explicit QA checkpoints.*

Figure 1 provides a high-level view of the end-to-end data pipeline, organized into six stages covering data extraction, integration, cleaning, enrichment, analysis, and visualization. Each numbered box corresponds to a notebook that produces a versioned staging dataset, supporting traceability and modular validation.

Figure 2 complements this view by showing the external data sources, the APIs, and the generated artifacts per integration point, clarifying how inputs enter the pipeline and how datasets flow across environments.

Stages 1 and 2 construct the film–album spine from TMDB and MusicBrainz data; **Stages 3–4** handle cleaning, canonicalization, and enrichment; and **Stages 5–6** perform exploratory analysis, statistical testing, and visualization on the finalized analytics datasets.

2.1 TMDB spine

The TMDB (The Movie Database) is a community-built database providing extensive information on movies. Key features of this dataset include financial performance metrics (budget, revenue), production details (studios, director, top cast), musical credits (soundtrack, composer), and audience engagement metrics (popularity, vote count, rating). TheTMDB dataset serves as the "spine" of our analysis; its metadata provides numeric and categorical attributes about the movie itself and allows it to be linked to secondary datasets to obtain additional attributes about the soundtracks. Using the TMDB API, we retrieved 31,978 films from 2015-2025 (to reduce junk entries, we had pulled entries with a vote count > 10).

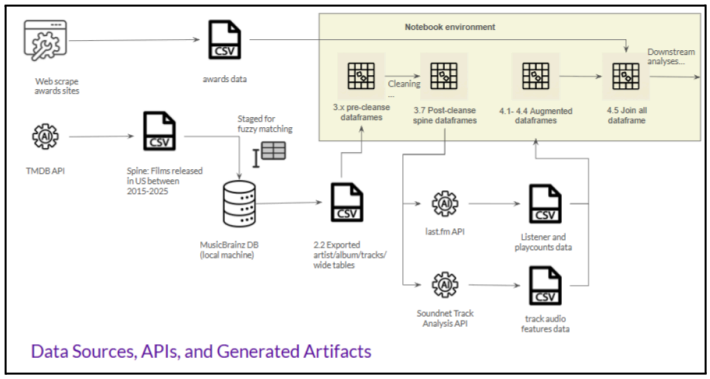

Figure 2. *External data sources and generated artifacts across the analysis pipeline.*

2.2 MusicBrainz spine

MusicBrainz models albums as a collection of normalized entities (release groups, releases, labels, tags, and relationships), requiring multiple joins to reconstruct an album-level view suitable for analysis. It's pretty massive: about 141GB!

2.2.1 Album data

The entity-relationship diagram (ERD) in **Figure 3** simplifies the key relationships within MusicBrainz that make up the concept of an "album", whose closest approximation in MusicBrainz is called a release group. We won't go into details here, but the most important points to realize are:

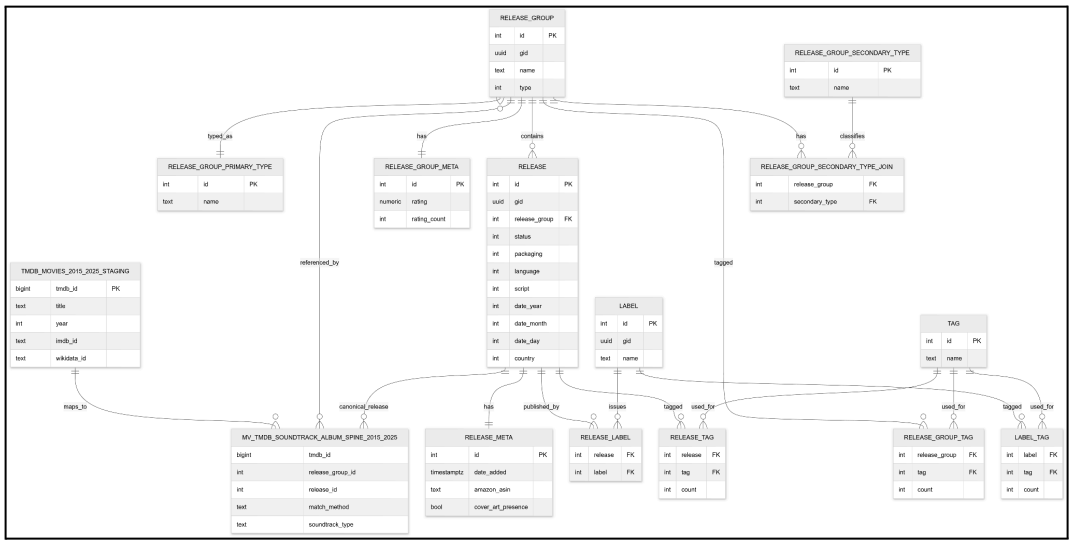

Figure 3. *ERD of album-level entities and join with imported and staged TMDB data*

- The SQL notebooks included **are** reproducible provided you download and install the MusicBrainz DB on a Docker instance in your local machine (~12 hours).
- Release groups consist of releases in multiple countries. Our analysis is focused on the United States, so we limit information to releases there.
- The release group secondary type fortunately marks "Soundtracks", so it serves as a useful feature in limiting albums pulled
- TMDB_Movies_2015_2025_staging imports the CSV from Notebook 1.1 and joins it with MusicBrainz data using two approaches:
  - IMDb ID matching, which yields a 4.4% match rate when IDs align across TMDB and MusicBrainz.
  - Heuristic title matching (e.g., Film: "The King" → Album: "Official Soundtrack of The King"), which increases the match rate to 15.0% but introduces false positives addressed in Notebook 3.3. We also apply strict duration controls, requiring album release dates (when available) to fall within three years of the film's release.
- The resulting materialized view, MV_TMDB_SOUNDTRACK_ALBUM_SPINE_2015_2025, which has 5209 records, forms the final film–album spine and is serialized to CSV for downstream Python-based cleansing and analysis.

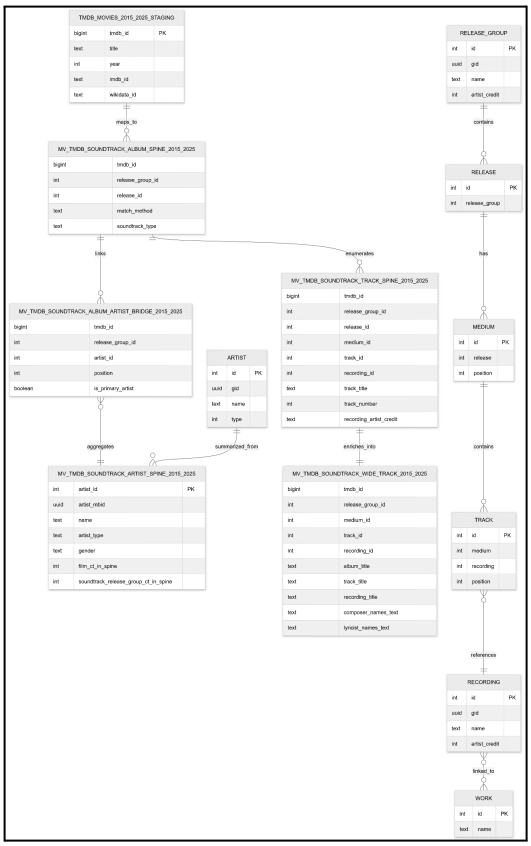

Figure 4. *ERD for supporting entities*

2.2.2 Other data

Figure 4 shows a simplified ERD of our final materialized views and supporting tables for artists, tracks, and the wide dataset. We created an album_artist bridge to model the many-to-many relationship between soundtracks and artists, which supports composer data reconciliation. Track-level data also incorporates attributes from related entities such as recording and work.

All supporting materialized views are derived exclusively from albums defined in the film–album view (Section 2.2.1), ensuring consistent scope and preventing unintended row expansion.

2.3 Secondary sources

Secondary sources are joined only **after** the spine is cleansed, to prevent unnecessary API data pulls. We export 2430 artist records, and 85K rows. Wide also has 85K rows, but contains the combined rows from all three base Materialized Views (91 columns total).

2.3.1 Awards data

To analyze the "prestige factor" of movie soundtracks, a data pipeline was developed to aggregate award data from the Oscars, Golden Globes, Critics' Choice, and BAFTAs. Due to the lack of programmatic access on official award websites, a BeautifulSoup scraper was used to extract records from Wikipedia. To ensure high match rates between music datasets and award lists, a robust normalization engine was built to standardize titles by removing accents, stripping "The" prefixes, and converting Roman numerals. The final data was transformed into a machine-ready format using Boolean flags for 15 specific categories, identifying both nominees and winners. This process was executed across four granular levels—Artists, Albums, Tracks, and Wide files—and verified through a custom QA script to confirm that certain films were correctly captured.

2.3.2 last.fm data

Last.fm provides raw listening counts, which we found more interpretable than Spotify's popularity scores. The data represent a point-in-time snapshot collected between Jan 29 and Feb 3, 2026, including listeners (unique users) and playcount (total listens) at the artist, album, and track levels.

Last.fm metrics are incremented depending on how each entity is accessed: listeners who listen by accessing an artist or album page increment the corresponding artist and album metrics (along with track metrics for the tracks they listen to), while listens originating from playlists increment only track-level metrics. These measures are used as outcome variables for correlation analysis, with the caveat that Last.fm data reflects a self-selecting user population (selection bias), and that more recent albums have had less time to accumulate listening activity (survivorship bias), which can favor older releases.

2.3.3 Audio features

In order to obtain audio features for individual tracks, we took the artist names and track titles from the MusicBrainz dataset (with minor cleaning for apostrophes rendering as other symbols), obtained the Spotify Track IDs via the Spotify Developer API, then queried the Soundnet Track Analysis API on RapidAPI to obtain the features, such as tempo (beats per minute), musical key, and happiness and danceability scores (scale of 0 - 100). The APIs returned results in JSON, which were then converted to CSVs. The timeframe was the same as the MusicBrainz data extraction (2015 to 2025). Due to significant slowdowns and recurring errors with the Soundnet API, we narrowed the population to tracks that belonged to films with a film_vote_count >= 500 votes. After applying this filter, we queried the API for 33,247 tracks that had valid Spotify IDs. The success rate in obtaining usable features from the API was ~97%.

III. Data manipulation and cleaning

3.1 Spine manipulation and cleanup

Stage 3 (**Figure 5**) focuses on cleaning and manipulating the project spine. We export several materialized views from the local MusicBrainz database for downstream preprocessing.

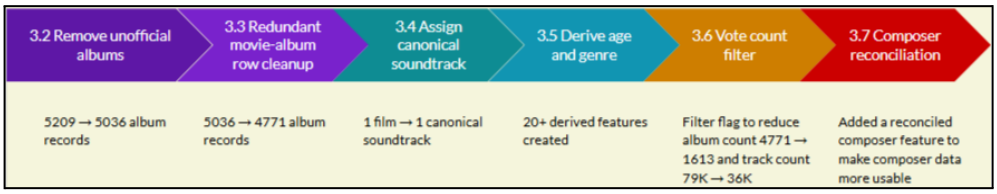

Figure 5. *Overview of the spine pre-processing*

Notebook 3.1 QA Gate: Spine integrity checks. We verified primary key integrity, assessed categorical coverage, and flagged sparsely populated attributes. All key integrity checks passed. Several attributes (e.g., work-level composer and lyricist) were identified as too sparse to be useful, and these findings informed the Stage 3 cleanup roadmap.

Notebook 3.2 Remove unofficial albums. Some albums were marked as *Bootleg* or *Promotional* via release_status. To maintain a clean spine, we restricted the data to *Official* releases only, reducing the album count from 5,209 to 5,036.

Notebook 3.3 Redundant movie-album row cleanup. Heuristic album-name matching in Stage 2 produced 448 duplicate albums, identified via repeated release_group_ids. We resolved these redundancies through a multi-pass heuristic that prioritized film–album name consistency, external identifier matches (IMDb), and film popularity (TMDB vote count), ultimately retaining a single album per tmdb_id. All deletions were propagated to the track and wide tables. Most removed records corresponded to short or ambiguous film titles (e.g., Alone, Gold, Drive), which are especially prone to name collisions. This step reduced the album set from 5,036 to 4,771 records.

Notebook 3.4 Assign canonical soundtrack. To prevent films with multiple soundtrack releases from disproportionately influencing results, we designate a single canonical album per film. Rules are applied sequentially: films with one album retain it; when both a score and soundtrack exist, the score is selected and the compilation labeled as a songtrack; remaining ties are resolved using IMDb identifiers and deterministic ordering. This ensures each film contributes at most one album to downstream analyses.

Notebook 3.5 Derive age and genre. We compute film and album age (in days) using release dates and a reference timestamp of Jan 31, 2026. Genre metadata is moderately sparse (≈20% coverage) and highly granular, with over 350 community-defined terms. To make genre analytically tractable, we map these terms into seven higher-level buckets: classical/orchestral, electronic, ambient/experimental, rock, pop, hip hop/R&B, and world/folk. Film genres are retained as provided by TMDB, while artist genres are similarly bucketed for completeness. All genre variables are represented as boolean flags.

Notebook 3.6 Vote count filter. Due to significant slowdowns with the Soundnet API, we prioritized films with a vote count greater than or equal to 500. We performed analysis on well-known composers (e.g., John Williams, Hans Zimmer) to ensure that this filter wasn't too restrictive. We also inspected a sample of excluded films as a sanity check. Prioritizing these movies introduces selection bias, as users are more likely to leave ratings for well-known films. This could cause the under-representation of smaller, less commercially successful movies in our analysis. This step reduced the dataset from 78,992 to 36,580 records.

Notebook 3.7 Composer reconciliation. The composer reconciliation process resolves discrepancies in TMDB data—such as missing entries (i.e. unknown 16.7%) or unranked multiple composers (16.6%)—by cross-validating it against a MusicBrainz "truth map." The script attempts to find an exact string match between both sources; if successful, it selects the verified MusicBrainz name (ranked). If no match is found, it follows a strict fallback hierarchy: prioritizing the first TMDB name, then the first MusicBrainz name, and finally defaulting to "Unknown." This unified result is then merged into the main dataset to ensure consistency across all analysis files.

Notebook 3.8 QA gate: Post-cleanse sanity check. After spine cleansing, no duplicate (tmdb_id, release_group_id) pairs remain, all required fields are non-null, and each film is associated with exactly one canonical soundtrack. Genre metadata is available for approximately 25% of albums, while composer data is present for roughly 98%. Applying the vote-count filter at a later stage reduces the album set from 4,771 to 1,791 and the film set from 4,448 to 1,622. Manual spot checks

across prestige, franchise, and independent composers confirm that the vast majority of U.S. films remain represented after pruning.

3.2 Post–secondary source manipulation and cleanup

In parallel with spine cleanup, we extracted and joined secondary data sources. Following the full merge (Notebook 4.5), we conducted additional post-join QA and exploratory checks to validate consistency before analysis.

Notebook 4.6 Raw metrics QA. We were able to retrieve last.fm listener metrics for over 90% of our album dataset (95% if filtered to vote count > 500). Track retrieval rates are similar.

Both album- and track-level last.fm metrics exhibit heavy right skew and kurtosis, motivating log transformation of listener and playcount metrics. This is especially evident when plotting histograms of log album listeners and log track playcounts, where the underlying distributional shape becomes clearer (**Figure 6**).

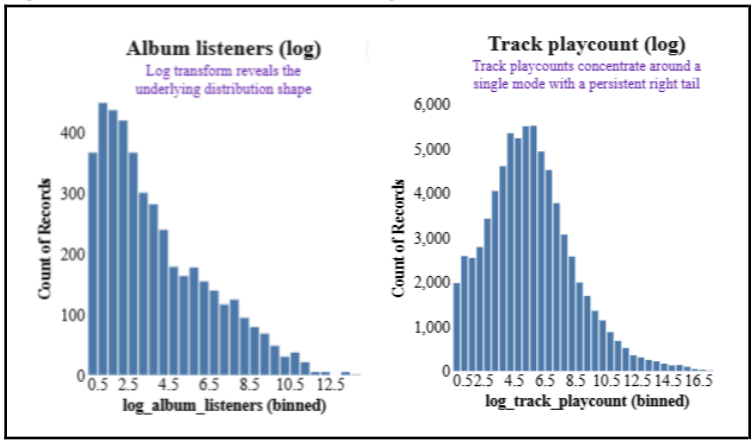

Figure 6. *Shape of distribution emerges after log-transforming metrics*

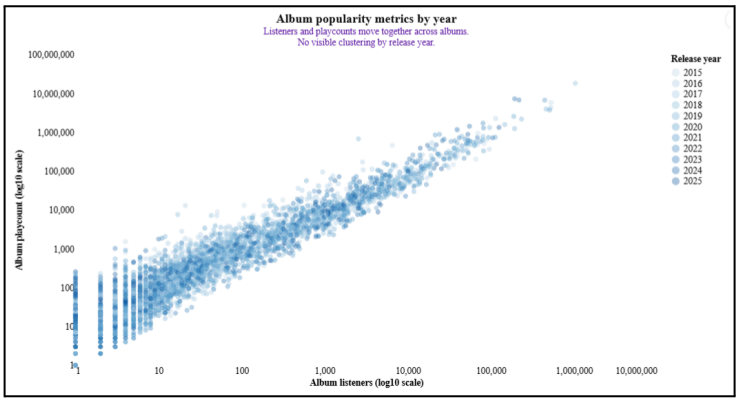

Figure 7. *Album listeners (x) plotted against album playcount (y) on logarithmic scales, showing a tight monotonic relationship.*

Album listeners and album playcounts move closely together, consistent with a very strong monotonic relationship (**Figure 7**; Spearman $r \approx 0.96$). The same pattern holds at the track level (Spearman $r \approx 0.97$). Given this tight alignment, we simplify downstream analysis by focusing on one metric per entity: album listeners as a measure of reach, and track playcounts as a proxy for repeated listening behavior.

In contrast, album-level and track-level popularity are only moderately aligned. Album listeners and the most-listened track show a Spearman correlation of 0.39, and track-level metrics often exceed album-level counts by large margins. This reinforces that album and track metrics capture different aspects of popularity and should be analyzed separately.

Notebook 4.7. Transform and reduce to an analytic set. We log-transform all listener and playcount metrics and reduce each of our dataframes to our analytics set (finally removing all records associated with film vote count < 500), and write it to file. These dataframes will be the basis of all analysis and visual exploration in Phases 5 and 6.

Notebook 5.4a. Audio feature cleaning. After obtaining the results from the Spotify and Soundnet APIs, we used track_title_cleaned, recording_artist_credit, and spotify_track_id to join the audio features with the wide and track datasets. After joining, we noticed some instances where the same Spotify ID was assigned to songs with similar, but clearly different track titles (for example, Spotify returned the same ID for tracks called "Space Gate" and "Assault at the Space Gate"). To solve this, we recovered matched_track_name from the API result files, normalized track names in both datasets, and tagged false positives. We set the audio features to "NA" to prevent inaccurate data from feeding into our analysis. Including records that encountered API errors, we had 3,366 records with null or missing values. In the interest of maintaining variation in the feature columns for our analysis, we decided to drop these from the dataset.

IV. Analysis and visualizations

4.1 Film/Album exploratory

A key question is whether critical recognition, film and soundtrack popularity align with commercial success. **Figure 8** Q-Q plot demonstrates that Film Revenue may be skewed in their raw form showing deviations from the normality line.

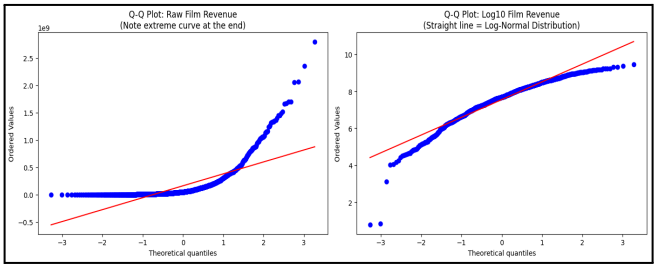

Figure 8. *Quantile - Quantile Plot to verify data is normally distributed*

Applying a Log10 transformation linearizes the data, as evidenced by the points closely following the reference line in red. This transformation supports the use of medians as stable measures of central tendency, forming the basis for our quadrant analysis.

Figure 9 plots revenue (log scale) against film popularity (log scale). Award winners (violet) and nominees (red) cluster primarily in the upper quadrants. This suggests that, from 2015 to 2025, major award nominations are associated with films above the overall revenue baseline — a commercial proxy for prestige — with nominations clustered in the top half and wins occurring almost exclusively in the upper-right quadrant (higher film popularity).

Circle size denotes soundtrack popularity via album playcounts. Although wins concentrate in the upper-right quadrant, nominations are more dispersed, with a meaningful share appearing in the lower-left quadrant (below-median revenue and popularity).

Overall, award bodies appear internally consistent in their win criteria. Films in the lower-right quadrant (higher popularity, below-median revenue) may reflect cult-style recognition, where audience engagement outpaces box-office performance despite limited awards success.

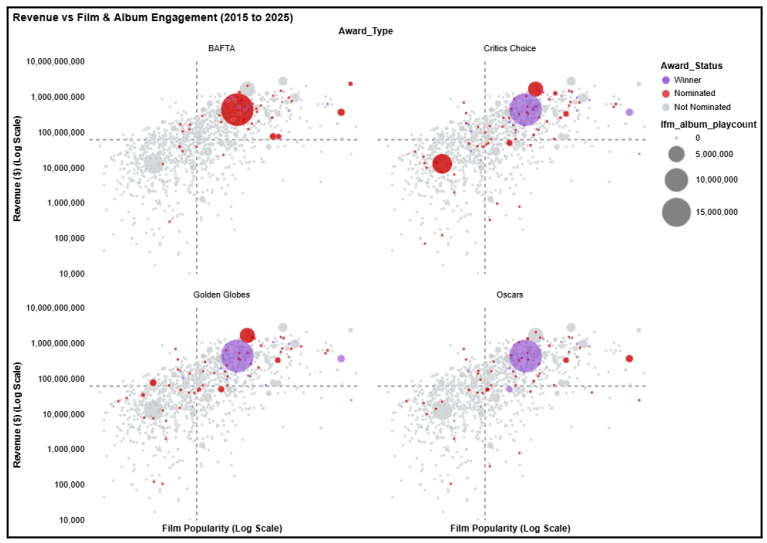

Figure 9. *Film Revenue plotted against film popularity (y) on logarithmic scales, showing the prestige factor*

4.2 Last.fm metrics exploratory

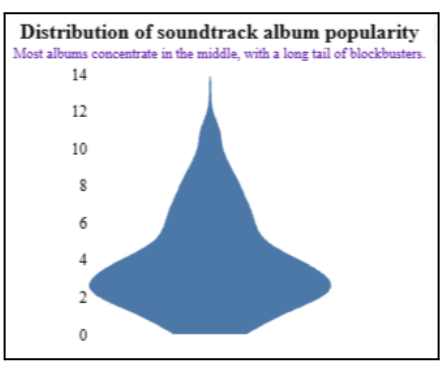

Figure 10. *Listener counts cluster centrally, with a long tail of blockbusters.*

We examine the distribution of soundtrack popularity using album listeners. **Figure 10** shows most albums cluster within a narrow listener range, with a small subset forming a long upper tail. The violin plot highlights this central concentration and the rare blockbusters that drive the spread.

To understand how popularity is distributed within albums, we examine the share of total track-level listening captured by an album's most popular tracks using Top-1 and Top-3 listening share distributions. **Figure 11** shows that soundtrack listening is typically concentrated rather than evenly distributed across tracks. While most albums avoid complete single-track dominance, listening attention is rarely uniform: for many albums, just one track, and more commonly the top two or three tracks, accounts for a large share of total listening.

To further illustrate how album- and track-level popularity relate, **Figure 12** plots album listeners against the most-listened track. While some albums lie near the parity line, many diverge substantially, with the top track attracting far more listeners than the album overall. The relationship is moderate rather than tight (Spearman r ≈ 0.39), with especially wide variation among lower-listener albums. Together, these patterns show that album-level and track-level metrics are related but not interchangeable.

Album listeners capture broad interest in a soundtrack, while track-level metrics reflect which specific songs listeners gravitate toward. As a result, albums with similar overall popularity can exhibit very different internal listening structures, reinforcing that album- and track-level metrics are complementary, not interchangeable.

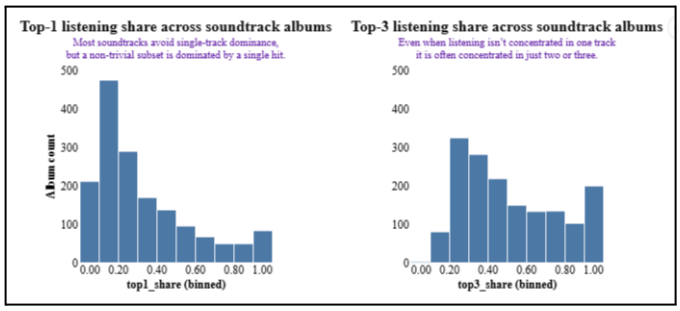

Figure 11. *Listening often concentrates in a few tracks despite limited single-track dominance*

4.3 Audio features exploration

We begin by exploring track tempo, measured in beats per minute (BPM) in **Figure 13**. One team member has experience as a DJ and music producer, bringing domain knowledge and perspective to this topic. Tempo is critical to a DJ's craft as it determines what songs pair well together for transitions during a performance. We hypothesized that soundtracks, which primarily feature orchestral/classical music, would be significantly slower than more upbeat genres, such as electronic dance (EDM) music. The tempo distribution for soundtracks appears bimodal, with peaks near ~90 and ~130 BPM. Contrary to our hypothesis, this overlaps with typical EDM tempo (~120–130 BPM).

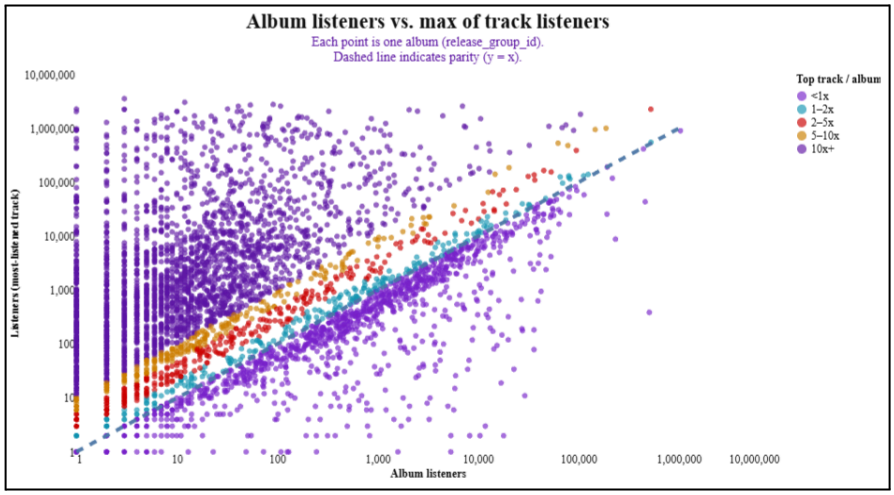

Figure 12. *Each point represents one album; deviations from the parity line show that track-level reach often diverges substantially from album popularity.*

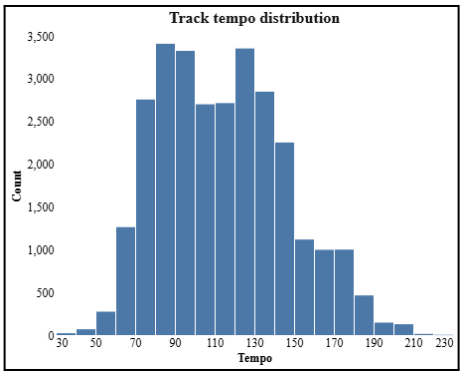

Figure 13. *Histogram of tempo (beats per minute) across all tracks.*

Via outside research, we also learned that the median tempo in our dataset (111 BPM) was consistent with the allegretto pace in classical music (112 - 120) [8]. Visualizing tempo allowed us to identify a data quality issue where some songs were registering 0 BPM. Likely an error in the Soundnet algorithm, we dropped these 39 records before plotting.

Next, we looked at happiness scores (1 - 100) in **Figure 14**. Our hypothesis was that tracks associated with movie soundtracks would tend to be more emotionally heavy (as opposed to light and airy) in order to evoke more powerful feelings from the audience. In music theory, songs with a major mode are perceived as happier, and those with minor sadder. We expected that the happiness scores from Soundnet would at least partially be driven by this well-accepted indicator. However, in the correlation matrix that we discuss below, we found that the happiness score had no significant correlation with the mode of a song.

Further, our initial histogram showed that the happiness score was heavily right-skewed, with a considerable peak near 0. This seemed a bit extreme, and as we couldn't find definitive documentation from Soundnet on how this was derived (see notebook 5.4b for further discussion), we decided to rely on mode to quantify happiness in our analysis.

We created two views of mode: one for the top 7 composers by track play count **(Figure 15)**, and another for 6 film genres **(Figure 16)**. Contrary to our hypothesis, both composer and genre views show a majority of tracks in major mode (in other words, happier-sounding songs were favored).

For example, Alan Silvestri (of *Avengers* fame), had 67% of his tracks in major, and 33% in minor. In terms of genres, family films had the highest share of major mode tracks (67%), while drama and action share the lowest proportion at 62%. Unsurprisingly, comedy has more happy sounding tracks than drama at 66%.

Finally, we generated a correlation matrix **(Figure 17)** between the audio features and track play count, and found no significant correlation between these (using both the Pearson and Spearman methods). This suggests that other factors, such as artist or movie recognition, may be more significant drivers of play counts at the track level. The matrix also sheds light on how Soundnet may derive its happiness score. As we stated before, the lack of relationship between mode and happiness is noteworthy. However, we see energy, danceability, instrumentalness, and loudness scores all correlate significantly with happiness, suggesting that these are key drivers of the score.

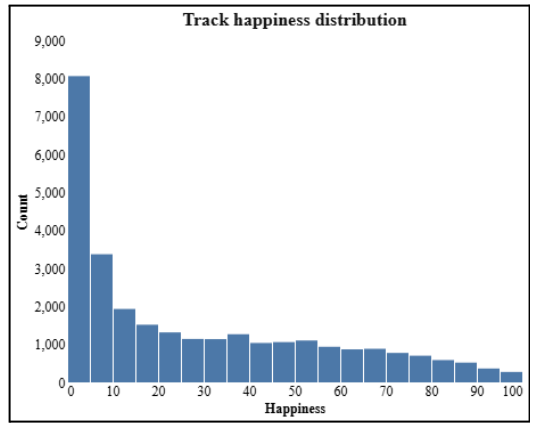

Figure 14. *Histogram of Soundnet's happiness score (1-100) across all tracks.*

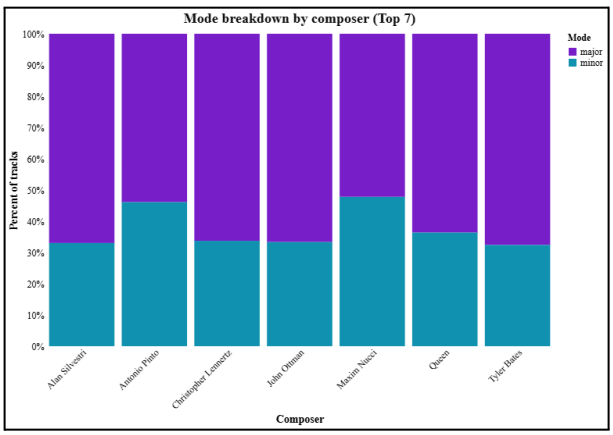

Figure 15. *Major/minor mode share among top 7 composers*

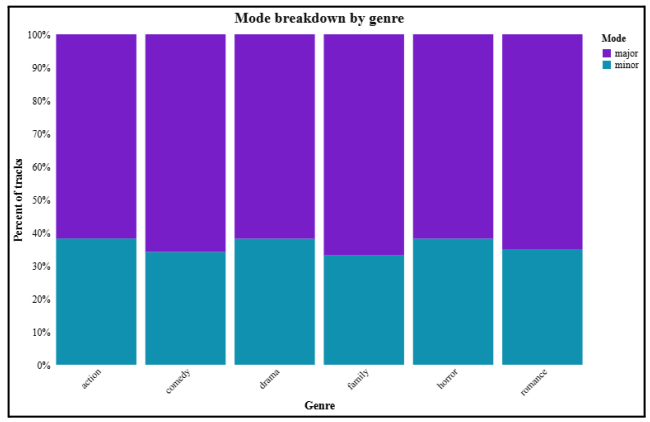

Figure 16. *Major/minor mode share by film genres of interest*

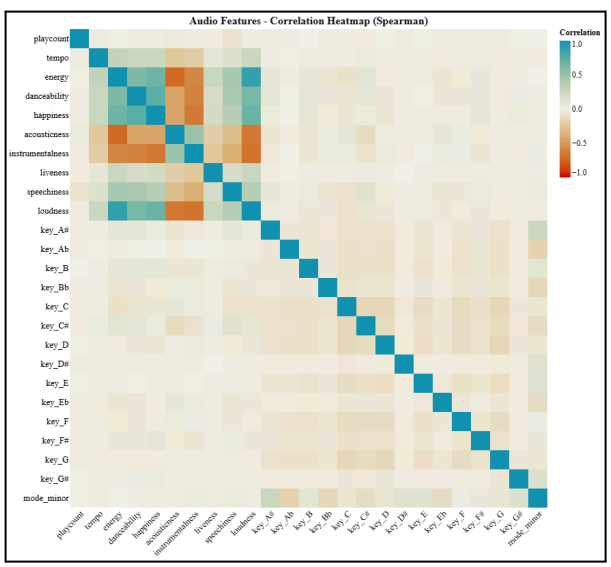

Figure 17. *Correlation heatmap between audio features and log track play count on Last.fm.*

4.4 Correlation and regression analysis

In this section, we address the question *What drives album listenership?* We examine relationships between album listeners and a broad set of film- and album-level features using correlation and regression analysis.

As a first step, we use a Spearman correlation heatmap (**Figure 18**) to identify clusters of highly correlated features. Specifically, film_vote_count, film_revenue, film_budget, and film_popularity form a correlated cluster, and we eliminate all save film_vote_count as the representative feature for regression to reduce the risk of multicollinearity. In parallel, we compute pairwise Pearson correlations (each feature x log album listeners) and rank them by the strength of their linear association, visualizing the results in a lollipop chart (**Figure 19**). The top 15 features identified by Spearman's & Pearson's correlations fully overlap, indicating that both methods surface a consistent set of candidate predictors.

We also visualize listener distributions across a select set of binary and binarized feature splits with a ridge plot (**Figure 20**). This lets us assess whether changes in feature values correspond to visible rightward shifts in the distribution of log album listeners. The clearest right shifts are award nominations, film exposure (aka film_vote_count) and album age above median. Others show weaker marginal separation despite modest pairwise correlations.

We then estimate a multivariate regression model using the culled featureset. Some attributes with weak pairwise correlation nevertheless emerge as strong predictors when considered jointly. For example, *film_is_documentary* (β = 2.5) and *film_is_animation* (β =0.87) have modest Pearson correlations but sizable positive effects in the regression. In contrast, *film_vote_count* (β = 0.60) and *classical_orchestral* albums (β = 0.57) perform strongly in both correlation and the multivariate model.

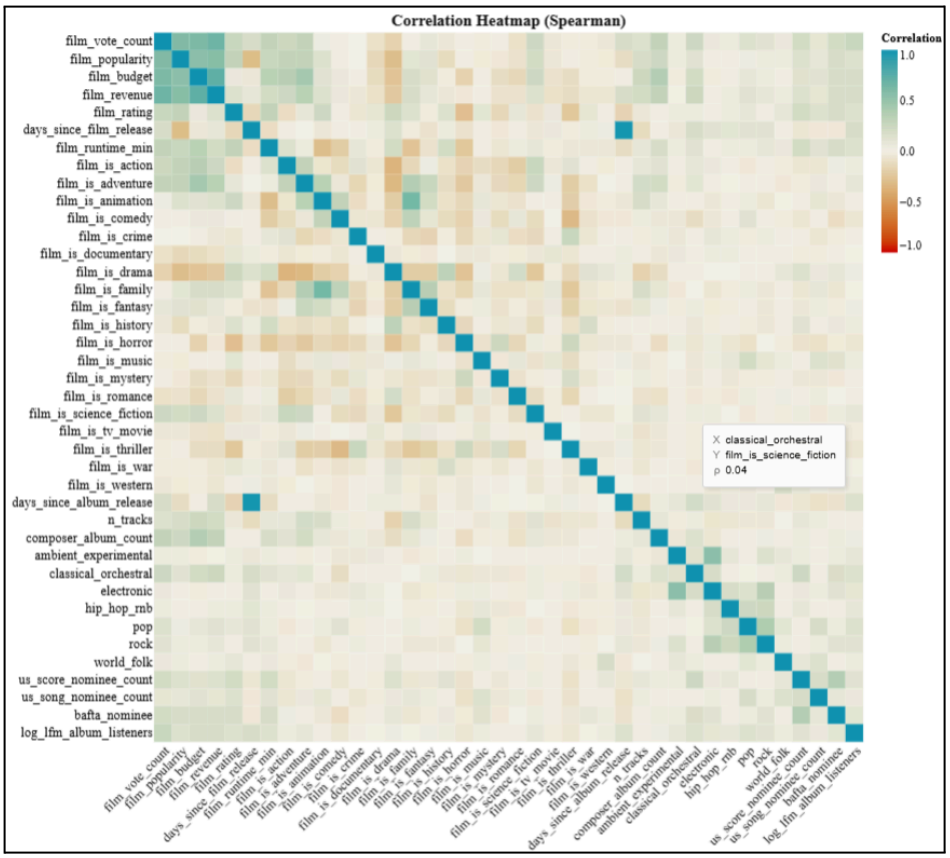

Figure 18. *Spearman correlation heatmap to detect multicollinearity before regression*

These four are the only predictors with positive coefficients that remain statistically significant at the 95% level, out of 30+ features. Regression coefficients and their 95% confidence intervals are shown in **Figure 21**.

The differences across Pearson, the ridge plot, and regression reflect the contrast of looking at features in isolation vs. evaluating them jointly. Some variables that show distinct rightward shifts in the ridge plot or modest pairwise correlations in the lollipop see those effects weaken once an exposure-related variable is included in regression. This tells us that the individual feature effect may actually be tied to some shared variance with film vote count or commercial scale. In other words, not every marginal signal survives when controls are introduced.

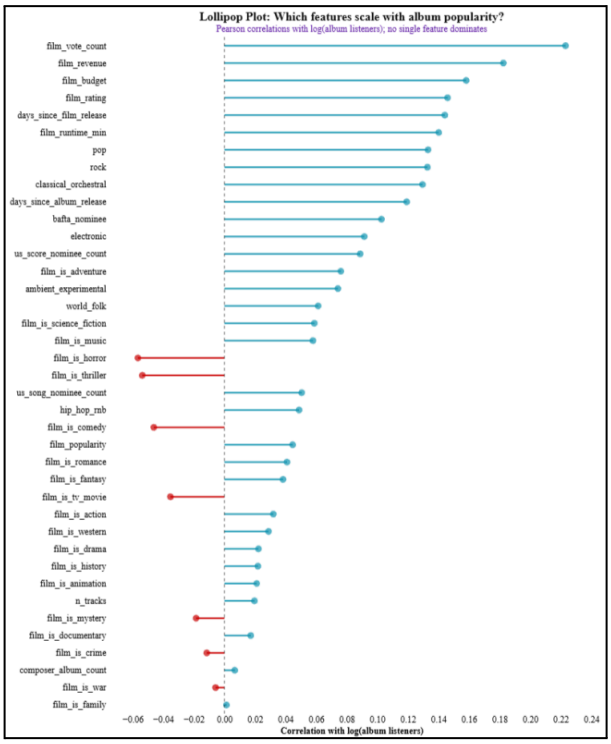

Figure 19. *Pearson correlations between features and log album listeners, ranked by magnitude*

V. Conclusion

The computational narrative that emerged from our 2015–2025 soundtrack analysis gives us a few key areas for further exploration regarding what drives listenership in the modern, disruptive streaming era. For instance, a soundtrack's

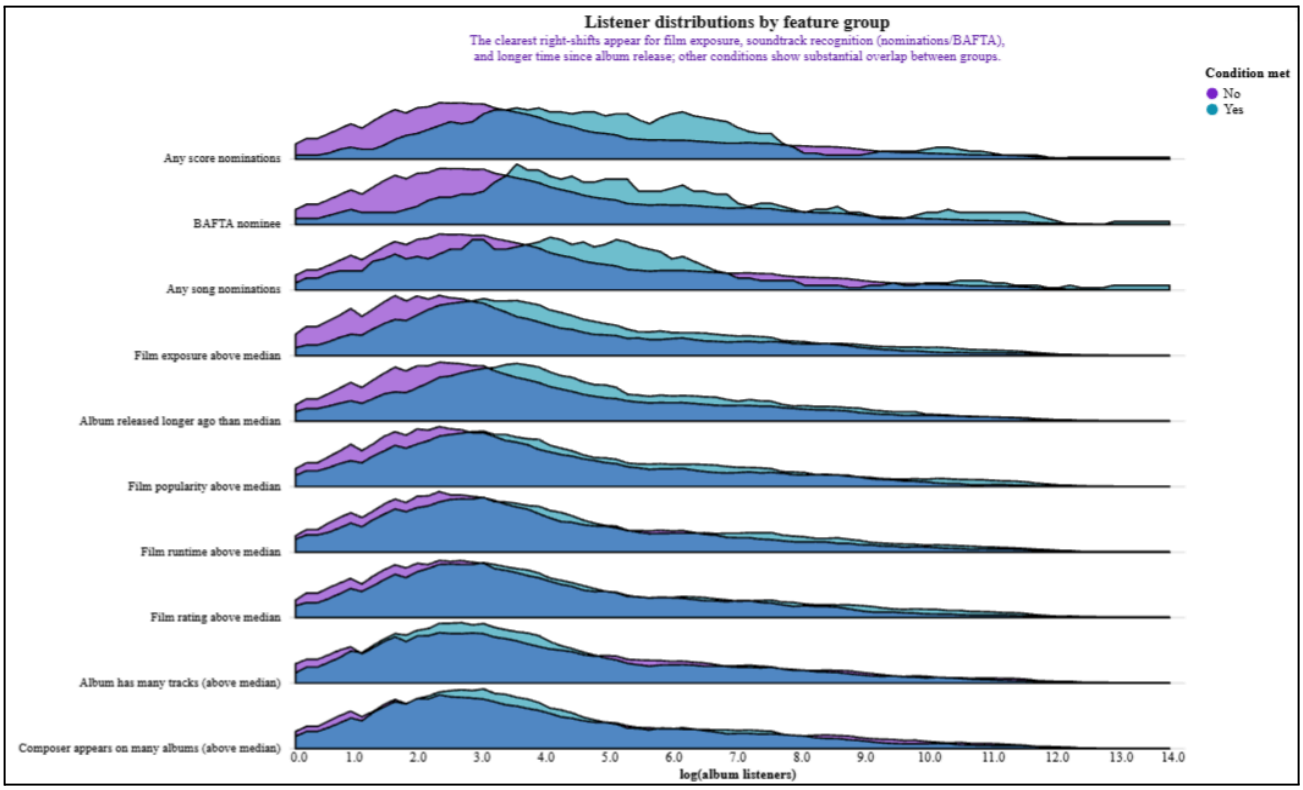

Figure 20. *Listener distributions by feature group (ridge plot), showing shifts in log(album listeners) across binary splits.*

popularity is moderately correlated with the commercial and critical success of the film itself.

The study found that films with high revenue and high popularity scores are strongly associated with soundtrack listener counts. Critical recognition and industry awards (Oscars, Golden Globes, BAFTAs, etc.) are highly concentrated among high-grossing, high-popularity films, while nominations extend further down the distribution, including a meaningful share among below-median films.

Soundtrack listening is rarely uniform across an album. Instead, engagement is typically concentrated in just one or three hit tracks that attract significantly more listeners than the album as a whole. Track-level audio features such as tempo, happiness, and danceability showed no significant association with listening outcomes.

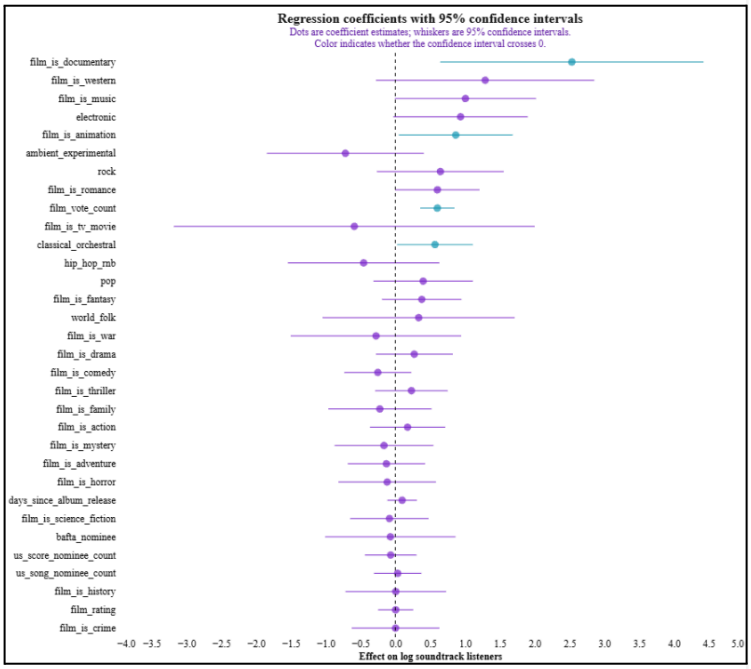

Figure 21. *Regression coefficients (95% CIs) of features that remain predictive of album listeners after controls*

While film popularity, film vote count, and revenue are highly correlated measures of commercial scale; specific genres also emerge as significant predictors of soundtrack listenership. After controlling for exposure-related variables, documentary and animated films, along with classical/orchestral albums, remain positively associated with higher streaming numbers. Many apparent predictors weaken once exposure-related variables are introduced, suggesting that much of the variation in soundtrack streaming is tied to underlying film visibility rather than intrinsic musical characteristics.

Appendix: Statement of Work + References

This project was divided based on each member's primary interests and desire to deepen a particular skill. While responsibilities were clearly delineated, all members contributed to ideation, proposal development, peer review, and final report writing. Major decisions—such as feature selection, matching logic, and modeling approach—were discussed and agreed upon collaboratively.

A.1. Contributions

Khurram Abbas: Led film-focused ideation; conducted TMDB exploration, filtering, and API extraction; performed composer reconciliation and cleansing; researched and scraped awards data; conducted film–listener analysis and related viz; contributed to report writing, managed certification and export from Deepnote to submission

Tony Lett: Led music-focused ideation and project coordination; researched audio features; implemented Spotify and RapidAPI extraction, cleanup, and merge; led track-level and audio-feature statistical analysis and viz; contributed to report writing.

Jaime Singson: Led metrics-based ideation and overall pipeline architecture; installed and explored MusicBrainz; implemented the film–album join and SQL-based data spine (extraction, cleanup, reconciliation, QA gates, canonization); led Last.fm API extraction and merge; conducted album-level statistical analysis and viz; contributed to report writing.

A.2. Collaboration Process

Collaboration occurred primarily through Slack, with two to three SCRUM-style meetings per week to review progress, knowledge-transfer dataset structure, resolve issues, and share each others' findings. Weekly milestones helped us maintain our momentum, while the pipeline visualization enabled parallelized workstreams. We met weekly with our mentor (Oleg) to refine the team's analytical direction.

Collaboration was effective due to high responsiveness, clear milestone setting, and shared visibility into dependencies. However, we underestimated API-related delays and credential access; in future work, we would secure access earlier and build more buffers into the timeline. We also leveraged AI tools to accelerate documentation, QA and as a brainstorming partner for more complex analysis.

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